ChE422 Biochemical Engineering

2020 Midterm Exam

3/10/2020

2:30 PM - 3:45 PM

Name: [redacted]

75%
10

Instructions: This is a close-book and close-note exam.

MULTIPLE CHOICE QUESTIONS (Part I):

Note: Only ONE answer is correct. Circle the correct answer. (2 pts for each problem)

1. The bond between a phosphate group and the ribose sugar group in RNA is called which of the following?
   (A) Phosphodiester linkage
   (B) Glycosidic bond
   (C) Peptide bond
   (D) Amide bond

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2. Which of the following plays a role in DNA transcription?
   (A) DNA polymerase
   (B) RNA polymerase
   (C) DNAase I
   (D) Guanyl transferase

-2

3. What signals the end of transcription?
   (A) Stop codon
   (B) Terminator
   (C) The end of the DNA chain
   (D) RNA polymerase runs out

-2

4. The start codon is:
   (A) AUU (B) UAA (C) AUG (D) UAG

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5. Which of the following is not a component of a nucleotide?
   (A) Phosphate group
   (B) Anti-codon
   (C) Ribose sugar
   (D) Nitrogen base

-2

6. What provides the energy that drives the addition of nucleotides to a growing DNA chain during replication?
   (A) The hydrolysis of peptides
   (B) The hydrolysis of the diphosphate group in dNTPs
   (C) The hydrolysis of UTP
   (D) The hydrolysis of GTP

√

7. Which of the following is the sequence of the 10 region of prokaryotic promoters?

(A) TATAAT (B) TTGACA (C) TATA (D) CAA

8. How many base pairs make up a codon?

(A) 1 (B) 2 (C) 3 (D) 4

9. How many amino acids are there?

(A) 20 (B) 30 (C) 40 (D) 50

-2

10. How many hydrogen bonds are formed between one A:T base pair?

(A) 1 (B) 2 (C) 3 (D) 4

-2

MULTIPLE SELECTION QUESTIONS (Part II):

Note: ONE or MORE answers can be correct. Circle all correct answers. (3 pt for each problem)

11. Which of the following are the protein secondary structures?

(A) Hairpin (B) a-helix (C) b-Sheet (D) Peptide bond

12. Which of the following in a plasmid controls the copy number of this plasmid in the host?

(A) Antibiotic resistance gene

(B) Multiple cloning sites

(C) LacZ gene

(D) Origin of replication (Ori)

13. Which of the followings are required for Polymerase chain reaction (PCR)?

(A) DNA template (B) T4 ligase (C) DNA polymerase (Taq) (D) DNA primers

14. Which of the following methods are commonly used to transform plasmid into *E. coli* cells?

(A) Cell fusion

(B) Heat shock method

(C) Electroporation

(D) DNA injection by a needle

-1

15. Which of the followings are the key components in a general Operon?

(A) Promoter (B) Operator (C) phage gene (D) Structural genes

-1

16. Which of the following methods are used in random mutagenesis?

(A) Error prone PCR
(B) Chemical Deamination
(C) DNA shuffling
(D) *E. coli* XL1 Red method

-1

17. Agarose gel electrophoresis is able to separate linear DNA fragments with which of the following parameter?

(A) Charge (B) Size (C) Secondary structure (D) Conformation

18. In order to use pET expression system for protein expression, the host cell should provide with which of the following component for transcription?

(A) Taq DNA polymerase (B) T7 polymerase (C) protease (D) Histag

-3

19. Which of the following methods can be used to introduce the mutation at the specific sites in a gene?

(A) UV irradiation
(B) Cassette mutagenesis
(C) PCR site-directed mutagenesis
(D) RACHITT method

-3

20. Which of the following compound is used in pBAD expression system to induce protein expression?

(A) Glucose (B) IPTG (C) L-arabinose (D) Lactose

Brief answer the following questions

1. What is Central Dogma of Molecular Biology? (5pts)

DNA $\rightleftharpoons$ RNA $\longrightarrow$ Protein

Transcription Translation

2. List three key differences between prokaryote and eukaryote? (5pts)

① the eukaryote has the nucleer core that 有无核膜包被的细胞核 covered by membrane, while prokaryote doesn't have it.

② the eukaryote has larger size than prokaryote.

③ the mRNA must be transfered through the nuclear membrane if it wants to translation. while prokaryote doesn't need to do so.

3. Describe the general key components of an expression plasmid? (5pts)

Antibiotic resistance gene

Multiple cloning sites, a place have at most 20 restriction sites

LacZ gene

Origin of replication

4. Describe the key differences between blunt-end ligation and sticky end ligation for cloning? (5pts)

sticky-end ligation can be controlled, while blunt-end ligation can NOT be controlled

G⌐G G|G C|C

5. List at least two methods to confirm the insertion of a gene into the expression plasmid? (5pts)

① DNA sequencing.

② blue-white screening. If inserted, the intact structure is broken and make result become white. If not inserted, then become blue.

6. Briefly describe blue-whitee screen? (10pts)

$A \xrightarrow{\beta\text{-gal}} B + C$ B 半乳糖 C 葡萄糖

乳糖

Usually, β-gal has specific gene that have aitibiotic resistance and can be alive in the gel. After left reaction, the gene is intact and make the result became blue. If the plasmid is inserted and the structure is not intact, it will not generate final product and make the result white.

7. After you find an interesting gene in a new organism, briefly describe the procedures to let *E. coli* express this protein. (15pts)

① use target gene to make DNA Template and generate a large amount of DNA by using PCR Method.

② split the DNA into two stands with DNA polymerase, and then insert these target gene into plasmid.

③ inserted plasmid is transferred into E. coli cell and break the Circle into a line.

④ These line works as tRNA, and in ribosome, they will be encoded into amino acid and protein.